

CASE STUDY · 2026

Primăria

Municipal administration portfolio/pilot-stage case study

Status & implementation boundary — What this case study does and does not claim

Status: Primăria is public portfolio/pilot-stage software. No public deployment is currently claimed. Repository: <https://github.com/Bogdan0708/PrimatIA> The statutory tax engine runs behind 32-table forced row-level security with AES-256-GCM CNP encryption, a PatrimVen XML exporter with golden-file tests (F3001–F3003, F3101), a GDPR erasure-request flow (request recording and notification cleanup; approval, anonymisation and deactivation are deferred), and a cited fiscal assistant with prompt-injection defence and PII redaction — 318 tests, CI. Legal, fiscal, security and ANAF-related code is implementation evidence only; it has not been independently certified as compliant and should not be read as legal assurance.

Role: Founder and sole engineer; Romanian municipal workflow concept and implementation.

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01 Product scope

Deterministic municipal workflow foundation

Checked-in surfaces

- Citizen and staff/admin interfaces in a Next.js PWA; demo tenant modelled on the Bogdan Vodă commune.
- Tax-domain calculation modules for buildings, land and vehicles; PatrimVen XML exporter with golden-file tests.
- Cited fiscal assistant with prompt-injection defence and PII redaction; GDPR erasure-request flow (request recording and notification cleanup; approval, anonymisation and deactivation deferred); document, reporting, authentication, tenant and audit-oriented modules.
- Romanian, Hungarian and English interface message catalogues.

Boundary

Availability of a module does not establish end-to-end municipal adoption, production hardening, legal correctness or third-party acceptance of generated exports.

02 Architecture

Matches the checked-in implementation

Primary stack

Next.js 14, TypeScript, React, PostgreSQL and Prisma form the primary application stack, with 32-table forced row-level security and AES-256-GCM encryption for CNP fields. Supporting code/configuration includes Redis/BullMQ, object storage, authentication, Zod validation, Vitest and Docker-oriented infrastructure.

The cited fiscal assistant's LLM calls are routed through AI-gateway (github.com/Bogdan0708/AI-gateway), a separately published multi-provider gateway.

AI-gateway is published as a six-provider OpenAI-compatible gateway with spend caps, request dedup, Prometheus/OpenTelemetry and a runtime response contract on the JSON health/providers/chat surfaces (streaming excluded); 89 tests, CI. Live: ai-gateway-382299704849.europe-west2.run.app/health responds 200 (v3.0.0) on Cloud Run; deployed revision not yet tied to the published source.

Design choices

- Keep tax calculations deterministic rather than delegating them to an LLM.
- Scope data and permissions by tenant and role with forced row-level security; treat it as a control requiring deployment validation, not a certification.
- Keep the cited fiscal assistant reviewable, with prompt-injection defence and PII redaction, and separate from authoritative calculations.

03 Evidence & next steps

Credible claims without certification language

What it proves

PrimärIA demonstrates a substantial TypeScript municipal-domain codebase, data modelling, deterministic tax logic, forced row-level security, a golden-file-tested PatrimVen exporter, a cited fiscal assistant and 318 tests in CI.

Remaining gates

- Independent legal and ANAF-format validation.
- Deployment hardening and anonymous availability checks before any hosted-service claim.
- Pilot evidence and measured user outcomes before adoption or impact claims.